

# Aakarsh Singh

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## Experience

### Co-founder & Design Engineer

Mare [🔗](#)

Remote  
2024 – Present

- Co-founded a reference tool for designers and artists and built it end to end — ingestion pipeline, embedding + clustering layer, and the Next.js/TypeScript interface; closed beta with 100+ creatives and 10,000+ reference items processed.
- Built the Python/FastAPI backend on Supabase (Postgres + pgvector) with a self-hosted embedding/LLM service on serverless GPU (Modal); deployed on GCP (Cloud Run, Cloud Tasks) with Stripe billing.

### XR Software Engineer

Date 0:0 - Artists Sara Niroobakhsh & Wafaa Bilal

Remote  
August 2025 – December 2025

- Developed a Unity and Meta XR SDK based MR/VR app, optimizing 3D scanned and artist-generated assets to reduce scene memory usage by 60% and achieve a stable 90 FPS on standalone Quest 3.
- Implemented real-time NoSQL synchronization across concurrent gallery-installation devices with persistent environment states.

### Fullstack Software Engineer

Callback

Tokyo, Japan/Remote  
May 2024 – December 2024

- Launched a Next.js, TypeScript, Express.js, and Firebase site with a user leaderboard, boosting weekly active users from 25 to 250 within three months on GCP Cloud Run.
- Built a scalable social REST API (user profiles and a friend graph) that doubled daily active users in a month.
- Developed an enterprise dashboard with Stripe payouts and automated weekly Twilio reports and data visualizations, growing active partners on the dashboard by 200%.

### XR Software Research Intern

NYU Tandon @ The Yard

New York City, USA  
June 2024 – August 2024

- Independently delivered optimized Unreal Engine environments, Niagara simulations, and motion/volumetric capture pipelines; showcased to 300+ participants at NYU's engineering research conference and ArtsIT: Game Creation Conference.

### XR Software Engineer Intern

NEEEU Spaces GmbH

Berlin, Germany  
May 2023 – July 2023

- Developed an AR ad campaign for BMW's lifestyle magazine, achieving a 20% increase in customer engagement.

## Education

### New York University

B.A. Interactive Media, minors in Computer Science, Sound-Music Computing, Art History

September 2021 – May 2025  
GPA: 3.92/4.0

- Coursework: Algorithms, Data Structures, Natural Language Processing, Shader Programming, VR/AR Development, Web App Development, Game Development.

## Selected Projects & Presentations

### LAND: Unreal Engine Film [🔗](#)

November 2024 – April 2025

- Experimental Unreal Engine film integrating fluid simulation, motion capture, photogrammetry, and custom-trained AI visuals; screened at Ars Electronica Festival, Linz (122,000+ festival attendees).

### Custom Video-Diffusion Pipeline — “I only experience things in latent space now.”

May 2026

- Built a video pipeline conditioning diffusion models on depth, optical flow, and attention to rework archival and found footage; screened at Shortwave x Soft Centre, Sydney Opera House (Vivid Live).

### Instructor – Architectural Association Visiting School, Warsaw [🔗](#)

July 2025

- Invited to teach game design and scripting at an international workshop in Warsaw for architecture and PhD students.

**Exhibitions & performances:** Ars Electronica (Linz) • Dark Mofo / Night Mass (Hobart) • Sydney Opera House / Vivid Live (Sydney) • MUTEK.AE (Dubai) • Jameel Arts Centre (Dubai) • Alserkal Avenue (Dubai) • Fuji Rock (Japan). Full list at [aakarsh.dev/cv](https://aakarsh.dev/cv) [🔗](#).

## Technologies

**Languages:** TypeScript, JavaScript, Python, C#, C++, SQL, GLSL/HLSL

**Web/Backend:** Next.js, React, Tailwind CSS, Framer Motion, Three.js/WebGL, FastAPI, Node.js, Supabase, PostgreSQL + pgvector, GCP (Cloud Run, Cloud Tasks), Cloudflare R2, Stripe, Docker, Git, Figma

**Real-time/XR:** Unreal Engine, Unity, TouchDesigner, Blender, Cinema4D, Ableton